



Pranav Khare <khare.pranav@gmail.com>

Invitation to Review for IEEE Access Access-2024-22456

1 message

IEEE Access <onbehalf@manuscriptcentral.com>

Wed, Jul 17, 2024 at 11:51 PM

Reply-To: ieeeaccess2@ieee.org

To: khare.pranav@gmail.com

18-Jul-2024

Dear Mr. Khare:

The manuscript entitled "Optimizing Product Variety for Balancing Market Share and Complexity Cost in Product Family Design" has been submitted to IEEE Access. Since your expertise is relevant to this article, we would be grateful if you would consider reviewing it.

The abstract appears at the end of this letter, along with the names of the authors. Please let me know within 2 days if you will be able to accept my invitation to review. If you agree to review, we ask that you provide your review comments within **7 days** to assist IEEE Access in maintaining its expedited peer review process of 4 to 6 weeks. If you are interested in reviewing but need additional time to complete the review, we can grant a reasonable deadline extension upon request. For additional information on IEEE Access and tips on completing a quality review, please visit our [Reviewer Best Practices guide](#).

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Thank you for your assistance with the peer review of this IEEE Access article.

Sincerely,

Prof. Derek Abbott

Editor-in-Chief, IEEE Access
ieeeaccess2@ieee.org

MANUSCRIPT DETAILS

TYPE: Research Article

TITLE: Optimizing Product Variety for Balancing Market Share and Complexity Cost in Product Family Design

AUTHORS: Oh, Kwansuk; Hong, Yoo Suk

ABSTRACT: Product variety has a complex impact on both market share and complexity cost, so determining an optimal variety is a difficult task for a manufacturing company. This paper proposes a variety optimization model that reflects the relationships among market, design, and production units considering the increasing tendencies of market share (concavity) and complexity cost (convexity). Optimization model for finding an optimal variety consists of two main parts: a demand model and a complexity cost model. A demand model is constructed with the nested logit model showing the concavely increasing tendency of variety impact on market share, and a complexity cost model is developed by adopting the zero-based costing approach in which complexity cost is measured by incremental cost depending on the addition of variants. In the case study, we applied a front chassis module family of an automobile to the optimization model by analyzing data from the Korean and European markets. The results show that the greater the similarity level, the perceived similarity of a company's products, the better it is to provide less variety. Also, the more flexible a company's production facilities are, the more diverse the products it can produce.

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